Reasoning and Writing Formally

A guide for upper-division mathematics students

Extracted from O. Kasum, *Uvod u matematiku* (2026)

Abstract

This guide describes a standard of mathematical writing under which every sentence of a proof can be matched to the rule of inference that licenses it, the shape of every proof is dictated by the logical form of its goal, and no object is named before its existence and uniqueness have been established. It sets out the rules, the proof skeleton attached to each logical symbol, the discipline governing definitions, and a checklist to be run before any proof is handed in.

Contents

1 What This Guide Asks of You

Most of the mathematics you have written so far has been convincing. This guide asks you to make it checkable: written so that a reader who trusts nothing can put a finger on each sentence and say which rule licenses it. That reader might be a sceptical grader, a proof assistant, a collaborator in another field, or yourself in six months.

The method has three parts.

a. A fixed, small set of inference rules, one or two per logical symbol. A proof is nothing but a sequence of applications of these rules and of results already proved.

b. A proof skeleton for every logical shape. The outermost symbol of what you are proving tells you how the proof begins, what its parts are called, and how it ends. You do not invent the structure; you read it off.

c. A discipline for definitions. You may not name a thing until you have proved that it exists and is unique. You may not write $\{x : P(x)\}$ until you have proved that P defines a set.

Everything else in this guide (the conventions, the vocabulary) exists to serve these three.

2 Deductions

Terminology 2.1. A statement is a sentence which is either true or false. A predicate $P(x)$ is a statement depending on a parameter x; it becomes a statement when x is given a value. □

Terminology 2.2. A deduction is a sentence of the form

"Suppose $P_1, P_2, \dots, P_n$. Then Q."

The statements $P_1, \dots, P_n$ are its hypotheses, and Q is its conclusion. The case $n = 0$ is allowed. □

Remark 2.3. Besides the explicit hypotheses there are implicit ones, the axioms, which are present in every deduction and are never listed. Only axioms enjoy this privilege; anything else that is assumed must appear in the list. □

Terminology 2.4. A deduction is accepted only once it has been justified. The justification is called a proof. Proofs are the only scientific method mathematics has. □

Remark 2.5 (The lemma principle). Once a deduction has been justified, it may be used in every later proof without being justified again. A justified deduction is called a fact, a lemma, or a theorem. Logically these words mean the same thing. They differ only in what the author is signalling: a lemma is a stepping-stone, a theorem is something of value in itself, a fact is neutral. Do not let the label do any logical work. □

Remark 2.6 (Two kinds of rule). Proofs are built from rules of inference. There are *basic* rules, accepted without discussion, and *derived* rules, namely every fact, lemma and theorem already justified. Each logical symbol comes with rules of two kinds:

a. a rule for *proving* a statement whose outermost symbol it is (an introduction rule);

b. a rule for *using* a statement whose outermost symbol it is (an elimination rule).

This gives a mechanical first move at every point of every proof: look at the goal and apply its introduction rule; look at each hypothesis and know its elimination rule. □

3 The Rules

The one rule that belongs to no symbol is the following.

Rule (Assumption). At any moment we may conclude any of our hypotheses, or any axiom.

In practice this rule is applied silently. When a proof is being written out in full detail, it is named at the point where a hypothesis is first invoked.

3.1 Connectives

Let P, Q and R be statements.

Rule (Proving conjunction). Having proved P and having proved Q, we may conclude $P \wedge Q$.

Rule (Using conjunction, left and right). Having proved $P \wedge Q$, we may conclude P; we may also conclude Q.

Rule (Proving disjunction, left and right). Having proved P, we may conclude $P \vee Q$; having proved Q, we may conclude $P \vee Q$.

Rule (Using disjunction: proof by cases). Suppose we have proved $P \vee Q$; that we have given a subproof which additionally assumes P and proves R; and that we have given a subproof which additionally assumes Q and proves R. Then we may conclude R.

Rule (Proving negation). Suppose we have given a subproof which additionally assumes P and proves $\perp$. Then we may conclude $\neg P$ in the main proof.

Rule (Using negation). Having proved $\neg P$ and having proved P, we may conclude $\perp$.

Rule (Proving implication). Suppose we have given a subproof which additionally assumes P and proves Q. Then we may conclude $P \Rightarrow Q$ in the main proof.

Rule (Using implication: modus ponens). Having proved $P \Rightarrow Q$ and having proved P, we may conclude Q.

Rule (Proving equivalence). Having proved $P \Rightarrow Q$ and having proved $Q \Rightarrow P$, we may conclude $P \Leftrightarrow Q$.

Rule (Using equivalence, direct and converse). Having proved $P \Leftrightarrow Q$ and P, we may conclude Q; having proved $P \Leftrightarrow Q$ and Q, we may conclude P.

Rule (Proving tautology). At any moment we may conclude $\top$.

Rule (Using contradiction: ex falso quodlibet). Having proved $\perp$, we may conclude any statement.

There is exactly one rule which is neither an introduction rule nor an elimination rule.

Rule (Classical contradiction: reductio ad absurdum). Suppose we have given a subproof which additionally assumes $\neg P$ and proves $\perp$. Then we may conclude P in the main proof.

Remark 3.1. Note the asymmetry. The rule for proving negation *adds* a negation: from a subproof $P \vdash \perp$ we get $\neg P$. Reductio *removes* one: from $\neg P \vdash \perp$ we get P. Without reductio one cannot get P from $\neg\neg P$, cannot prove $P \vee \neg P$, and cannot prove $\neg \forall x\, P(x) \Rightarrow \exists x\, \neg P(x)$. Every time you write "suppose, for contradiction, that not P" you are using reductio. Say so, and make sure you actually needed it. $\square$

3.2 Equality

Let a and b be objects and let $P(x)$ be a predicate.

Rule (Proving equality). At any moment we may conclude $a = a$.

Rule (Using equality: substitution). Having proved $a = b$ and having proved $P(a)$, we may conclude $P(b)$.

Remark 3.2. That is all equality is. Symmetry, transitivity and "replace a by b inside any expression" are all derived, by choosing the predicate well. For instance, to obtain $b = a$ from $a = b$, let $Q(x)$ be the predicate "$x = a$". Then $Q(a)$ holds by the rule for proving equality, and substituting $a = b$ yields $Q(b)$, that is, $b = a$. When you replace one thing by an equal thing inside a complicated formula you are applying the substitution rule; if it is not clear *which* occurrence is being replaced, name the predicate. □

3.3 Quantifiers

Let $P(x)$ be a predicate. The variable x in $\forall x\, P(x)$, $\exists x\, P(x)$ and $\exists! x\, P(x)$ is bound: the statement is not about x, and x may be renamed.

Rule (Proving the universal quantifier). Having proved $P(x)$ for an arbitrary given x, we may conclude $\forall x\, P(x)$.

Rule (Using the universal quantifier: instantiation). Having proved $\forall x\, P(x)$, and a being any object, we may conclude $P(a)$.

Rule (Proving the existential quantifier). Having proved $P(a)$ for some object a, we may conclude $\exists x\, P(x)$.

Rule (Using the existential quantifier: witness). Suppose we have proved $\exists x\, P(x)$, and that we have given a subproof which introduces a *new* name a, assumes $P(a)$, and proves a statement Q *which does not mention a*. Then we may conclude Q in the main proof.

Rule (Proving the uniqueness quantifier). Having proved $\exists x\, P(x)$ and having proved

$$\forall x\, \forall y\, (P(x) \land P(y) \Rightarrow x = y),$$

we may conclude $\exists! x\, P(x)$.

Rule (Using the uniqueness quantifier). Having proved $\exists! x\, P(x)$, we may conclude $\exists x\, P(x)$. Having proved $\exists! x\, P(x)$, $P(a)$ and $P(b)$, we may conclude $a = b$.

Remark 3.3. The two side conditions on the witness rule are where wrong proofs hide. The name a must be fresh, not already in use for anything else. Whatever is concluded about the outside world must not contain a. The sentence "Let a be such that $P(a)$" opens the subproof; the subproof ends when something has been reached that no longer mentions a. □

Remark 3.4. The "uniqueness" half of the rule for proving $\exists!$ establishes *at most one*, not *exactly one*. Existence supplies the rest. □

Notation 3.5. Let S be a set.

a. $(\forall x \in S)\, P(x)$ means $\forall x\, (x \in S \Rightarrow P(x))$;

b. $(\exists x \in S)\, P(x)$ means $\exists x\, (x \in S \wedge P(x))$;

c. $(\exists! x \in S)\, P(x)$ means $\exists! x\, (x \in S \wedge P(x))$. $\square$

Remark 3.6. Bounded quantifiers are abbreviations, and you should know what they abbreviate. "Let $x \in S$ be given" is really "let x be given; suppose $x \in S$". Exhibiting a witness for $(\exists x \in S)\, P(x)$ means exhibiting a with *both* $a \in S$ and $P(a)$. A quantifier with nothing after it (a bare "$\exists x$") is not a statement; to say that something exists, say $\exists x\, \top$. $\square$

3.4 Derived Rules Cited by Name

Each of the following is a few lines from the basic rules. Once proved, they are cited by name or number and never re-derived.

Fact 3.7. *Let P, Q be statements and let $P(x), Q(x)$ be predicates. Then:*

a. *(modus tollens) if $P \Rightarrow Q$ and $\neg Q$, then $\neg P$;*

b. *(contraposition) $(\neg P \Rightarrow \neg Q) \Leftrightarrow (Q \Rightarrow P)$;*

c. *(double negation) $P \Leftrightarrow \neg\neg P$;*

d. *(De Morgan) $\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$ and $\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$;*

e. *(De Morgan) $\neg \forall x\, P(x) \Leftrightarrow \exists x\, \neg P(x)$ and $\neg \exists x\, P(x) \Leftrightarrow \forall x\, \neg P(x)$;*

f. *$(P \Rightarrow Q) \Leftrightarrow (\neg P \vee Q)$ and $\neg(P \Rightarrow Q) \Leftrightarrow P \wedge \neg Q$;*

g. *if P does not mention x, then $\forall x\, (P \vee Q(x)) \Leftrightarrow P \vee \forall x\, Q(x)$, and likewise for $\wedge$, for $\exists$, and for the consequent of $\Rightarrow$;*

h. *if P does not mention x, then $\forall x\, (Q(x) \Rightarrow P) \Leftrightarrow (\exists x\, Q(x) \Rightarrow P)$;*

i. *$\forall x\, (P(x) \wedge Q(x)) \Leftrightarrow (\forall x\, P(x) \wedge \forall x\, Q(x))$ and $\exists x\, (P(x) \vee Q(x)) \Leftrightarrow (\exists x\, P(x) \vee \exists x\, Q(x))$;*

j. *$\forall x\, P(x) \vee \forall x\, Q(x) \Rightarrow \forall x\, (P(x) \vee Q(x))$ and $\exists x\, (P(x) \wedge Q(x)) \Rightarrow \exists x\, P(x) \wedge \exists x\, Q(x)$, and neither converse holds in general.*

Proof. Parts b, c, e and j require the classical rule; the rest do not. Each is a direct application of the skeletons of Section 4. We leave them as Homework 3.8. $\square$

Homework 3.8. *Prove each part of Fact 3.7, indicating where the classical rule is used.*

4 Reading the Proof Off the Goal

Here is the single most useful habit in this guide. Before writing a word of the argument, look at the outermost logical symbol of the statement to be proved, and open the corresponding skeleton. Then fill it in. Every skeleton has an opening sentence, named parts, and a closing sentence which states what was proved and by which rule.

4.1 The Skeletons

Goal $P \Rightarrow Q$.

> Suppose P.
> Our goal is to prove Q.
> [*Proof of* Q.]
> Hence, by the rule for proving implication, $P \Rightarrow Q$.

The sentence "our goal is to prove Q" is not decoration. In a proof three levels deep it is the only thing that stops writer and reader losing track of what is currently being shown.

Goal $\neg P$.

> Suppose P; we derive a contradiction.
> [*Proof of* $\bot$*: find some* Q *with* Q *and* $\neg Q$ *both available, and say which.*]
> Hence $\neg P$.

Goal P**, by reductio.**

> Suppose, for contradiction, that $\neg P$.
> [*Proof of* $\bot$.]
> Hence P, by the classical rule.

Prefer a direct skeleton whenever one exists. Reductio is the right tool when P is atomic, or when P is an existential statement with no constructible witness.

Goal $P \wedge Q$.

Two visibly separate subproofs, then: "Since P and Q hold, $P \wedge Q$."

Goal $P \vee Q$.

Either prove one side and conclude by the left or right rule, or, when neither side is available outright, reduce to an implication:

> By Fact 3.7.f it suffices to prove $\neg P \Rightarrow Q$.
> Suppose $\neg P$.
> [*Proof of* Q.]
> Hence $P \vee Q$.

Goal $P \Leftrightarrow Q$.

> We reason by double implication.
> **(⇒)** Suppose P. [...] Hence $P \Rightarrow Q$.
> **(⇐)** Suppose Q. [...] Hence $Q \Rightarrow P$.
> Hence $P \Leftrightarrow Q$.

The halves may be labelled **Direct implication.** and **Converse implication.** The order is immaterial. If one direction is a known result, write "(⇒) is Fact 3.7.c" and prove only the other.

Goal $\forall x\, P(x)$**.**

> Let x be given.
> [*Proof of* $P(x)$*, using nothing about* x *except what the bound, if any, provides.*]
> Hence, by the rule for proving the universal quantifier, $\forall x\, P(x)$.

Goal $\exists x\, P(x)$**.**

> Let a be [*explicit construction*].
> [*Proof of* $P(a)$.]
> Hence $\exists x\, P(x)$.

Goal $\exists! x\, P(x)$**.**

> **Existence.** [*Proof of* $\exists x\, P(x)$.]
> **Uniqueness.** Let x and y be such that $P(x)$ and $P(y)$. [*Proof of* $x = y$.]
> **Conclusion.** By the rule for proving the uniqueness quantifier, $\exists! x\, P(x)$.

Goal $A = B$ **for sets.**

Either **Direct inclusion.** / **Converse inclusion.**, or, when the two inclusions are mirror images:

> Let x be arbitrary.
> Then
>
> $$x \in A \Leftrightarrow [\ldots] \Leftrightarrow [\ldots] \Leftrightarrow x \in B,$$
>
> where the first equivalence holds by [...] and the second by [...].
> The conclusion follows by the axiom of extensionality.

Goal $f = g$ **for functions.**

> The functions f and g have the same domain, namely X, because [...].
> It remains to show $(\forall x \in X)(f(x) = g(x))$.
> Let $x \in X$ be given.
> [*Chain* $f(x) = \cdots = g(x)$*, each link justified.*]

This applies the criterion that two functions are equal if and only if they have the same domain and agree pointwise; cite it.

Goal $(\forall n \in \mathbb{N})\, P(n)$**, by induction.**

> Let $P(n)$ be the predicate "[*write it out*]".
> We prove $(\forall n \in \mathbb{N})\, P(n)$ by induction.
> **Base of induction.** [*Proof of* $P(0)$.]
> **Inductive step.** Suppose $n \in \mathbb{N}$ and $P(n)$; we prove $P(S(n))$. [*Proof.*]
> **Conclusion.** By induction, $(\forall n \in \mathbb{N})\, P(n)$.

Write out the predicate. An induction whose predicate is never displayed is the leading source of the question "what exactly was the inductive hypothesis?", and of the error of using a stronger hypothesis than the one being proved.

4.2 Using the Hypotheses

Each hypothesis, by its outermost symbol, tells you what it can do for you.

a. $P \wedge Q$: split it; you now have P and Q.

b. $P \vee Q$: open a case analysis.

> Since $P \vee Q$ holds, we prove R by cases.
> **Case 1.** Suppose P. [*Proof of* R.]
> **Case 2.** Suppose Q. [*Proof of* R.]
> By cases 1 and 2, R.

   Every case must end in the *same* R. The closing line names the cases. Cases may be labelled by their assumption (**Case** $x = n$.) instead of by number.

c. $P \Rightarrow Q$: it is waiting for P. Obtain P, then apply modus ponens.

d. $\neg P$: it is waiting for P. Obtain P, then $\perp$.

e. $\forall x\, P(x)$: instantiate at whatever object you need.

f. $\exists x\, P(x)$: "Let a be such that $P(a)$." Fresh name; conclusion must not mention a.

g. $\exists! x\, P(x)$: you get a witness, and you get the collapsing rule: any two objects satisfying P are equal.

h. $a = b$: substitute, into a predicate you could name.

4.3 Recursing

Every skeleton opens subgoals. Each subgoal is a new goal: read off its outermost symbol, open its skeleton, fill it in. When the goal is atomic (a membership, an equation, a named relation) reach it by eliminating hypotheses, unfolding a definition, or citing an earlier result. The proof is finished when every opened block has been closed.

4.4 Moving the Goal

Two sentences move the goal rather than the hypotheses.

a. "By Fact 3.7, it suffices to prove [*new goal*]." Legitimate only when the cited result is an implication or equivalence whose conclusion is the current goal.

b. "It remains to show [*subgoal*]." After part of a conjunction, or the domain half of a function equality, has been dealt with.

Both commit you to proving exactly the stated new goal next.

5 Justifying Every Step

5.1 What Counts as a Reason

Each sentence of a proof is one of the following, and a reader should be able to tell which.

a. A hypothesis (assumption rule).

b. A basic rule applied to statements visible above.

c. An earlier result, cited by name or number.

d. A definition unfolded: "by definition of $\subseteq$".

e. A reduction of the goal, with its licence named.

5.2 What Does Not Count

"Clearly." "Obviously." "It is easy to see." "One can show." "It is well known." "A routine argument shows." "The details are left to the reader." None of these is a reason. A gap dressed in one of these phrases is still a gap.

Three delegations *are* permitted, because they lose no information.

a. A one-rule step may go unnamed when the two sentences around it determine the rule. "Since $P \wedge Q$ holds, P" needs no citation of the left rule for using conjunction. This is the *only* sense in which a step may be "obvious".

b. "Similarly" or "analogously", with the substitution stated. "The converse inclusion is proved analogously, with the roles of A and B exchanged." If the analogous argument needs a *different* hypothesis, say which. If you cannot state the substitution, the argument is not analogous.

c. A claim marked as unproven. If you cannot prove a claim you need, for lack of time or of technique, state it precisely, mark it explicitly as unproven, and go on using it. The reader then knows exactly what the rest of your argument rests on, and can judge for themselves whether the claim is true. What is forbidden is not the missing proof but the missing mark.

Remark 5.1. The phrase "it is not hard to check that" appears in the source book fewer than a dozen times in 120 pages, and every time the check is a direct unwinding of a definition displayed in the previous line. Hold yourself to that frequency and that condition. □

5.3 Chains

In a chain of equalities, equivalences or inclusions, every link gets a reason. When each link is a definitional unfolding visible from the surrounding text, the chain may run inline:

$$f^{-1} = f^{-1} \circ \mathrm{Id}_Y = f^{-1} \circ (f \circ g) = (f^{-1} \circ f) \circ g = \mathrm{Id}_X \circ g = g.$$

Otherwise the chain is displayed with numbered lines and one reason per line:

$$\begin{aligned} \mathrm{dom}(g \circ f) &= f^{-1}[\mathrm{dom}(g)] && (1) \\ &= f^{-1}[Y] && (2) \\ &= X, && (3) \end{aligned}$$

where Eq. (1) is by the formula for the domain of a composition, Eq. (2) uses $g : Y \longrightarrow Z$, and Eq. (3) uses $f : X \longrightarrow Y$. A chain of six links with a single "by definition" at the end is a gap five links wide.

5.4 Closing What You Open

Every "Suppose", every "Let x be given", every "Case k", every "(⇒)" opens a block. Every block is closed by a sentence which says what it established and by which rule, before the main line resumes:

> Hence $\neg P$ (rule for proving negation).
> By cases 1 and 2, R.
> Hence $\forall x\, P(x)$.
> Hence $Q \Rightarrow P$.

A long structured proof ends with a line labelled **Conclusion.** restating the conclusion of the theorem.

5.5 Deep Nesting: Breadcrumbs

When blocks nest more than two deep, prefix each inner label with its path.

> **Successor step > existence.** [...]
> **Successor step > existence > case 1:** $\xi < \alpha$. [...]
> **Successor step > existence > case 2:** $\xi = \alpha$. [...]
> **Successor step > existence > conclusion.** $g(\xi) = H(g{\restriction}\xi)$.
> **Successor step > uniqueness.** [...]
> **Successor step > conclusion.** $P(\alpha)$.
> **Limit step.** [...]
> **Conclusion.** For every ordinal α there is a unique α-good function.

Every opened block gets a "> conclusion" line before its sibling opens. This keeps a four-deep transfinite argument readable in ordinary prose.

6 Definitions Are Licences

Three kinds of thing may be defined, and each has a principle of definition telling you what to prove first and what you get afterwards.

6.1 Constants

Rule (Principle of definition of a constant). Let $P(x)$ be a predicate and suppose $\exists! x\, P(x)$. Then a new name c may be introduced for the unique x satisfying $P(x)$. At any moment thereafter, $P(c)$ may be concluded.

The written form is always the pair

> **Fact.** There exists a unique X such that $\Phi(X)$.
> **Definition.** [*Name*] is the unique X such that $\Phi(X)$.

The fact comes immediately before the definition. It is never omitted and never postponed. Afterwards, using the definition means one of two things, and you say which: "by definition of c, $\Phi(c)$" (the principle), or "since $\Phi(a)$ also holds, $a = c$" (the collapsing rule for $\exists!$).

6.2 Operation Symbols

Rule (Principle of definition of a unary operation symbol). Let $P(x, z)$ be a predicate and suppose $\forall x\, \exists! z\, P(x, z)$. Then a new symbol f may be introduced such that, for every object x, $f(x)$ is the unique z with $P(x, z)$. For any object c, $P(c, f(c))$ may be concluded at any moment.

The binary case is the same, from $\forall x\, \forall y\, \exists! z\, P(x, y, z)$.

Remark 6.1. An operation symbol is *total*: it must have a value on every object. If the intended definition makes sense only for some inputs (ordinals, finite sequences, non-empty sets), one must either

a. assign a stated default to all remaining inputs, e.g. "$H(x) = 0$ whenever x is not an ordinal", and say in a remark that the default is the value given to inputs one does not care about; or

b. define a *function* (a set of pairs) on a specified domain instead, which is partial by nature.

Keep the distinction: an operation symbol is a rule defined everywhere; a function is a set with a domain. Restricting a rule F to a set X gives the function $\{(x, F(x)) : x \in X\}$; extending a function to a rule requires choosing a default outside its domain. □

6.3 Relation Symbols

Rule (Principle of definition of a relation symbol). Let $P(x, y)$ be a predicate. Then a new name N may be introduced such that, for all x and y, $x\,N\,y \Leftrightarrow P(x, y)$. From $x\,N\,y$ we may conclude $P(x, y)$, and conversely.

No existence, no uniqueness: a relation symbol is pure abbreviation. Write it with the biconditional displayed.

Definition 6.2. The relation symbol $\subseteq$ is defined as follows: for all x and y,

$$x \subseteq y \iff \forall z\,(z \in x \Rightarrow z \in y).$$

□

6.4 Set-Builder Notation

Definition 6.3. A predicate $P(x)$ defines a set iff there is a set X such that $\forall x\,(x \in X \Leftrightarrow P(x))$. □

Fact 6.4. *Suppose $P(x)$ defines a set. Then there is a unique set X such that $\forall x\,(x \in X \Leftrightarrow P(x))$.*

Proof. **Existence.** This is the hypothesis.

Uniqueness. Let X and Y be such that $\forall x\,(x \in X \Leftrightarrow P(x))$ and $\forall x\,(x \in Y \Leftrightarrow P(x))$. Let x be arbitrary. Then $x \in X \Leftrightarrow P(x) \Leftrightarrow x \in Y$, both equivalences by instantiating the hypotheses at x; hence $x \in X \Leftrightarrow x \in Y$ by transitivity of $\Leftrightarrow$. Hence $\forall x\,(x \in X \Leftrightarrow x \in Y)$, and $X = Y$ by the axiom of extensionality. □

Notation 6.5. Suppose $P(x)$ defines a set. Then $\{x : P(x)\}$ denotes the unique set X such that $\forall x\,(x \in X \Leftrightarrow P(x))$. □

Remark 6.6. Not every predicate defines a set: by Russell's argument, "$x \notin x$" does not, and neither does $\top$. So the moment you write $\{\ldots : \ldots\}$ you owe a proof that its predicate defines a set. The standard method is a *bounding set*: exhibit B with $\forall x\,(P(x) \Rightarrow x \in B)$; separation then yields $\{x \in B : P(x)\}$, and it is the set wanted. Each set-existence axiom is used exactly once, to show that one specific predicate defines a set, after which the corresponding notation is introduced:

Axiom	Predicate shown to define a set	Notation then introduced
Separation	$x \in A \wedge P(x)$	$\{x \in A : P(x)\}$; $\varnothing = \{x : \bot\}$
Pairing	$x = a \vee x = b$	$\{a, b\}$; $\{a\} = \{a, a\}$
Union	$(\exists A \in X)(x \in A)$	$\bigcup X$; $A \cup B = \bigcup\{A, B\}$
Power set	$x \subseteq S$	$\mathcal{P}(S)$
Replacement	$(\exists x \in A)(y = t(x))$	$\{t(x) : x \in A\}$
Infinity	(an inductive set exists)	$\mathbb{N}$, the least inductive set

Note that $\bigcap X$ is defined only for non-empty X: the bounding set is an element of X, so one needs one. □

6.5 Notation with a Side Condition

Some notations are meaningful only under a hypothesis. State the condition inside the notation's definition, and treat it as a proof obligation whenever the notation is used.

Notation 6.7. Let $t(x)$ be an expression and let A and B be sets. The expression $A \longrightarrow B : x \mapsto t(x)$ is defined only when $(\forall x \in A)(t(x) \in B)$, in which case it denotes the function $\{(a, t(a)) : a \in A\}$. □

Likewise f^{-1} is a function only when f is an injection; $\bigcap X$ exists only when $X \neq \emptyset$; "the least element" is available only after existence and uniqueness of least elements has been proved.

6.6 Definition, Terminology, Notation

Remark 6.8. Three kinds of stipulation, three jobs.

a. *Definition* introduces a new symbol under one of the principles above. It has content: the licensing fact.

b. *Terminology* says how an existing predicate or symbol is read or named: "the predicate $x \subseteq y$ is read: x is a subset of y"; "we call x a natural number if $x \in \mathbb{N}$". No new symbol, no proof; but citable.

c. *Notation* introduces an abbreviation: dropped parentheses (after associativity has been proved), bounded quantifiers, $f : A \longrightarrow B$, R^{-1}, Id_X. Every dropped parenthesis is backed by a stated convention ("$P \wedge Q \wedge R$ means $(P \wedge Q) \wedge R$") and, where regrouping is used, by a result saying that the other groupings are equivalent.

A definition that only renames is Terminology; a definition that only abbreviates is Notation; a definition proper always answers: *the unique what, satisfying what?* Logically all three are stipulations of the same kind, and nothing is lost if you do not keep them apart; the distinction is a courtesy to the reader, who then knows in advance whether a proof obligation is attached. □

6.7 Introducing a New Object, in Order

a. State the property $\Phi(X)$ the object should have, completely, with all quantifiers displayed.

b. Prove existence (typically: a predicate defines a set, or a witness is constructed).

c. Prove uniqueness (typically: extensionality; or the criterion for equality of functions; or the rule for using $\exists!$).

d. Write: *Definition.* [*Name*] is the unique X such that $\Phi(X)$.

e. Separately: Terminology (how it is read), Notation (abbreviations).

f. Prove the first properties as facts which unfold Φ, so that later proofs cite those and never return to Φ directly.

7 Writing Mathematics

7.1 Keep the Three Kinds of Text Apart

A mathematical text consists of three kinds of material, and the reader must be able to tell at a glance which is which.

a. *Statements*: definitions, terminology, notation, facts, lemmas, theorems. Each is stated completely, on its own, before anything is said about it.

b. *Proofs*: the justifications of the statements. A proof contains only steps of the kinds listed in Section 5.1.

c. *Commentary*: motivation, intuition, warnings, history, comparison with other approaches. Commentary is valuable, but it goes in remarks and examples, never inside a statement and never inside a proof.

A proof that pauses to explain why the theorem is interesting has stopped being a proof. Say it before, or say it after.

7.2 Stating Results

a. State the hypotheses first, then the conclusion. Do not scatter hypotheses through the conclusion ("… provided that …").

b. Every quantifier must be present, and preferably in front of the statement it governs. It may be written in symbols or in English: "for every x in the domain of f, $f(x)$ lies in the codomain of f" is as good as $(\forall x \in \operatorname{dom} f)(f(x) \in \operatorname{cod} f)$. What is not acceptable is a statement with the quantifier missing, or trailing so far behind that its scope is unclear: in "$f(x)$ lies in the codomain of f for every x in the domain" the reader must reconstruct what the "for every" governs. Whatever the notation, the scope of every quantifier must be unambiguous.

c. In a definition the biconditional is "iff"; in a result it is "if and only if". The word marks whether a sentence is a stipulation or a claim.

d. Refer to earlier results by name or number, never by "as shown above", "by the previous argument", "by the lemma". A reader who cannot find what you are citing cannot check what you are claiming.

7.3 Variables

a. Every variable in a statement is either bound by a displayed quantifier or introduced by "Let … be given" before the statement. A letter that is neither is a letter whose meaning the reader has to guess.

b. A variable introduced inside a block dies when the block closes. Do not use it outside without re-introducing it, and do not give a new object a name already in use for something else (this is the freshness condition on witnesses).

c. Bound variables may be renamed; fixed ones may not.

7.4 Unresolved Material

If a step is genuinely missing and you cannot fill it, mark it visibly, in the text or in the margin. If a whole claim is used without proof, state it as a claim and mark it as unproven. Never bridge a hole with a gesture. A marked hole is a debt; an unmarked one is a lie.

8 Worked Examples

8.1 A Propositional Identity

Fact 8.1. *Suppose* $\neg(P \vee Q)$*. Then* $\neg P \wedge \neg Q$*.*

Proof. We prove $\neg P$ and $\neg Q$ separately, and then combine them.

Suppose P; we derive a contradiction. Since P holds, $P \vee Q$ holds by the left rule for proving disjunction. Since $\neg(P \vee Q)$ and $P \vee Q$ both hold, we obtain $\perp$ by the rule for using negation. Hence $\neg P$, by the rule for proving negation.

Suppose now Q; we again derive a contradiction. Since Q holds, $P \vee Q$ holds by the right rule for proving disjunction. Since $\neg(P \vee Q)$ and $P \vee Q$ both hold, we obtain $\perp$ by the rule for using negation. Hence $\neg Q$, by the rule for proving negation.

Since $\neg P$ and $\neg Q$ hold, we conclude $\neg P \wedge \neg Q$ by the rule for proving conjunction. □

Remark 8.2. Every sentence of the preceding proof names the rule it applies and the statements it applies it to. In longer proofs the name of a basic rule may be dropped when the two surrounding sentences determine it; the name of a derived result never is. □

8.2 A Statement About Functions

We assume the standard facts about composition: if $f : X \longrightarrow Y$ and $g : Y \longrightarrow Z$, then $g \circ f : X \longrightarrow Z$ and $(\forall x \in X)((g \circ f)(x) = g(f(x)))$; we refer to this as the composition criterion.

Fact 8.3. *Suppose*

1. $f : X \longrightarrow Y$*,*
2. $g : Y \longrightarrow Z$*,*
3. $g \circ f$ *is an injection.*

Then f *is an injection.*

Proof. By definition of injection, our goal is

$$(\forall x_0 \in X)(\forall x_1 \in X)(f(x_0) = f(x_1) \Rightarrow x_0 = x_1).$$

Let $x_0, x_1 \in X$ be given and suppose $f(x_0) = f(x_1)$; we prove $x_0 = x_1$. Substituting $f(x_0) = f(x_1)$ into the predicate "$g(f(x_0)) = g(t)$", which holds at $t = f(x_0)$ by the rule for proving equality, we obtain $g(f(x_0)) = g(f(x_1))$. By the composition criterion, $(g \circ f)(x_0) = g(f(x_0))$ and $(g \circ f)(x_1) = g(f(x_1))$; hence, by transitivity of equality, $(g \circ f)(x_0) = (g \circ f)(x_1)$. By the composition criterion, $\mathrm{dom}(g \circ f) = X$, so $x_0, x_1 \in \mathrm{dom}(g \circ f)$. Since $g \circ f$ is an injection, we conclude $x_0 = x_1$.

Hence, by the rule for proving implication and then the rule for proving the universal quantifier applied twice, f is an injection. □

Remark 8.4. Every sentence of the preceding proof is one of the five things listed in Section 5.1. Compare the version one might have written: "If $f(x_0) = f(x_1)$ then clearly $g(f(x_0)) = g(f(x_1))$, so $x_0 = x_1$ since $g \circ f$ is injective." Same idea; three gaps: which rule gives the first equality, why $(g \circ f)(x) = g(f(x))$, and why x_0, x_1 lie in the domain of $g \circ f$. □

8.3 A Definition, Done in Order

Fact 8.5. *Let A and B be sets. Then there is a unique set X such that*

$$\forall x\, (x \in X \Leftrightarrow x \in A \wedge x \in B).$$

Proof. Let X be $\{x \in A : x \in B\}$; this is a set by the axiom of separation.

Existence. By the definition of $\{x \in A : \ldots\}$, X has the required property.

Uniqueness. Let X' also satisfy $\forall x\, (x \in X' \Leftrightarrow x \in A \wedge x \in B)$. Let x be arbitrary. Then

$$x \in X' \Leftrightarrow x \in A \wedge x \in B \Leftrightarrow x \in X,$$

where the first equivalence is the hypothesis on X' instantiated at x and the second is the definition of X. Hence $X' = X$ by the axiom of extensionality. □

Definition 8.6. Let A and B be sets. Then $A \cap B$ is the unique set X such that $\forall x\, (x \in X \Leftrightarrow x \in A \wedge x \in B)$. □

Terminology 8.7. The expression $A \cap B$ is read "the intersection of A and B". □

Homework 8.8. *Let A, B and C be sets. Prove that $(A \cap B) \cap C = A \cap (B \cap C)$.*

Notation 8.9. Let A, B and C be sets. Then $A \cap B \cap C$ denotes the set $(A \cap B) \cap C$. □

Remark 8.10. By Homework 8.8, $A \cap B \cap C$ is also equal to $A \cap (B \cap C)$. Nothing here is named before it exists; the notation with dropped parentheses is introduced only after the result which makes the grouping immaterial. □

9 Before You Hand It In

Run through the following. Each "no" is a gap, a compression, or an error; find out which.

The statement.

- Are all hypotheses stated? Is anything assumed in the proof that was not stated?
- Is the conclusion one fully quantified statement? Is every variable bound or introduced?
- Does every quantifier say what it ranges over and which statement it governs?

The skeleton.

- Does the opening of the proof match the outermost symbol of the goal?
- Is every "Suppose", "Let", "Case", "($\Rightarrow$)" closed by a sentence saying what it established?
- Does every case end in the same statement, and does the closing line name the cases?
- Is every "for contradiction" really reductio (goal P, assumption $\neg P$), and every "suppose P; we derive a contradiction" really the rule for proving negation (goal $\neg P$)?
- Does every induction display its predicate verbatim and use exactly that as the hypothesis?
- Are blocks deeper than two levels labelled with breadcrumbs?

The steps.

- Is each sentence a hypothesis, a basic rule on visible statements, a citation of an earlier result, a definition unfolding, or a licensed goal reduction?
- Is there any "clearly", "obviously", "easy to see", "one can show" standing in for a step?
- Does every link in every chain have a reason?
- Is every witness name fresh, and is the conclusion drawn free of it?
- Is every "Let x be given" x genuinely arbitrary?
- Is every use of $a = b$ a substitution into a nameable predicate?
- Does every modus ponens have both premises visible?

The definitions used.

- Does every constant have a preceding existence-and-uniqueness fact?
- Is every $\{\ldots : \ldots\}$ backed by a "defines a set" result, or of the form $\{x \in A : \ldots\}$?
- Is the condition of every side-conditioned notation discharged where it is used?

The delegations.

- Is every omitted proof either one rule, "analogously" with the substitution stated, or a claim explicitly marked as unproven?
- Is every step you could not justify marked rather than papered over?

The text.

- Is commentary out of the statements and proofs?
- Are hypotheses stated before the conclusion?
- Is every quantifier present, and is its scope unambiguous?
- Are all references to earlier results by name or number?

10 Why Bother

The standard described here is higher than most published mathematics meets. You will not always write at this level, and you should not always try: a research paper written for experts compresses on purpose. But you should always be *able* to. The skill of expanding any step of your own argument into named rules on demand is what lets you know, rather than believe, that your proof is right, and it is the only reliable way to find out when it is not. Compression is a choice made after the full argument is in hand. It is not a substitute for having it.

One more thing. Formal writing is not a translation step performed on finished thinking. Thinking formally means asking, of each step, which rule licenses it, what exactly the current goal is, and which hypotheses are available; writing formally means putting the answers on the page. The two activities are essentially indistinguishable, and this is why the discipline is worth acquiring: a person who can write a proof to this standard has thereby learnt to think one.

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